

SNAPFRAME · CALCULATION

SnapFrame sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-01	Amish Chadha	Design for construction (SNF-DDR-003, Draft) rerun; cable bolt check added; cost reported against the value-engineering target
0.3	2026-09-26	Amish Chadha	Budget approved by Amish (\$400 to \$445, SNF-DDR-002); script rerun; R10 not met to met
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). 1 in ridge tubes and one eave node variant; results rerun
0.1	2026-09-25	Amish Chadha	First sizing note for TRL 3 (geometry, wind, bracing, anchors, joints, mass, packages, cost, skin, erection time)

SnapFrame sizing calculations

With the decisions in SNF-DDR-001 and SNF-DDR-002 (1 in EMT rafters and ridge tubes, 3/4 in EMT posts and eave tubes, one eave node variant, tarpaulins from agency stock), the size M frame meets its floor, headroom, size-family and part-count targets, but five of thirteen requirements are **not met**: wind (R6), packages (R5), skin (R7), carried mass (R11) and fire (R13). A first-principles load share puts more wind load on the ridge tubes and posts than the TRL 2 estimate did. With the 1 in ridge tubes of SNF-DDR-002 the frame is rated at about **19.7 m/s** (71 km/h, 44 mph), up from 17.8 m/s in v0.1, and the 3/4 in posts now govern. The design for construction (SNF-DDR-003, version 0.4 of this note) adds cable bolt sets and second snap hooks: the frame kit is now about **\$469, \$24 over the \$445 value-engineering target** (a control target, not a limit; Amish, 2026-10-01), the tube bundle stays at **28.6 kg** against the 25 kg package limit, and the complete kit with tarpaulins is **52.5 kg**, 2.5 kg over the 50 kg carry limit.

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root). Geometry, cut lengths and node volumes come from the parametric model `cad/src/model.py`, so the note, the model, drawing SNF-DWG-001 and `bom/bom.csv` agree. This is a first-order hand calculation, not a code check or a frame analysis. Version 0.2 reruns the script after SNF-DDR-002; every changed number is noted with its v0.1 value. Version 0.3 changed only the budget, from \$400 to \$445 (SNF-DDR-002). Version 0.4 reruns the script on the constructable model of SNF-DDR-003: the hitch pin moves to 15 mm from the tube end, the cables run between their real attachment points (anchor eyes and cable bolt rings), the cable bolt is checked, and the masses and costs of the new parts are added. Every changed number gives its v0.3 value. The budget is reported as a value-engineering target, as Amish set out on 2026-10-01.

■ 1. Method

1. Build the size M geometry from the model parameters: 4.0 x 4.0 m floor, two 2.0 m bays, eave nodes at 1.8 m, ridge nodes at 2.6 m, foot node centers 60 mm above ground.
2. Compute section properties of 3/4 in and 1 in EMT from the nominal ANSI C80.3 diameters and walls.
3. Apply a gust dynamic pressure to the skin, share each tarpaulin panel's load between its four edge members with 45° tributary lines, and check each member as simply supported between node centers.
4. Follow the loads into the cable bracing, the foot anchors and the pinned joints.
5. Add up mass, package size and cost from the model and the BOM prices, and check the skin area and erection time.

■ 2. Assumptions

Table 1. Assumptions. Each is used only where stated.

#	Assumption	Value	Basis
A1	3/4 in EMT	OD 23.42 mm (0.922 in), wall 1.245 mm (0.049 in)	ANSI C80.3 nominal; OD and ID checked against Engineering ToolBox
A2	1 in EMT	OD 29.54 mm (1.163 in), wall 1.448 mm (0.057 in); rafters (SNF-DDR-001 D2) and ridge tubes (SNF-DDR-002 D8)	As A1
A3	Steel yield strength	275 MPa (40 ksi)	Assumed; C80.3 does not set a yield, so confirm from supplier data
A4	Design gust, safety factor	20 m/s, 1.5 on yield	R6
A5	Air density	1.225 kg/m ³	Sea level, 15 °C
A6	Pressure coefficients	Windward wall +0.8, leeward wall -0.5, roof -0.7 on both slopes, net of internal pressure for a closed shelter	First-order values in the style of ASCE/SEI 7; not a code check
A7	Open gable sensitivity	Internal pressure +0.55 when the wind blows into the open gable	Partially enclosed building value in the style of ASCE/SEI 7
A8	Panel load share	45° tributary lines from each panel corner: triangles on the short edges, trapezoids on the long edges	Usual two-way rule for a panel tied on four edges
A9	Members	Simply supported between node centers; node fixity ignored	Conservative for bending
A10	Bracing	Tension-only cables; the rear gable is the only braced frame across the span; the roof and side walls carry the rest to it	Concept bracing layout (SNF-PRC-001)
A11	Wire rope	4 mm 7x19 galvanized, minimum breaking load 8.0 kN	Typical catalog value, unverified
A12	Printed nodes	Density 1,070 kg/m ³ , 55 % effective fill, filament \$22/kg plus \$1.00 machine time per node	ASA-class polymer, estimate; polymer not chosen
A13	Tarpaulins	Two 4 x 6 m, 190 g/m ² , \$25 each, from agency stock	Relief tarpaulin class; outside the frame kit cost (SNF-DDR-001 D1)
A14	Tube prices	3.05 m (10 ft) stick: 3/4 in \$9.00, 1 in \$14.00	Indicative US retail, not quotes
A15	Corner eave nodes	The four-socket eave node with the socket past the gable left blank and capped	SNF-DDR-002 D9
A16	Cable bolt	M10 stainless A4-70 (yield 450 MPa), load applied at the ring 9.5 mm from the spot face, both cables of a rear corner eave node added at full value	SNF-DDR-003 C3; conservative upper bound
A17	Cable bolt sets and second snap hooks	0.13 kg and \$2.60 a set; cable assembly \$4.20 and 0.17 kg of fittings	Indicative catalog values

■ 3. Geometry

The size M floor is 4.0 x 4.0 m (16.0 m²), room for 4.6 people at the Sphere minimum of 3.5 m² each. The roof pitch is 21.8°. Taking the underside of the 1 in rafter, the roof is 2.0 m or higher over the central 2.92 m of the 4.0 m span, which is **73 %** of the floor (R2 needs 60 %).

Table 2. Cut lengths from the model (node-center distance less 45 mm at each end).

Size	Floor	Post	Rafter	Ridge and eave
S	3.0 x 3.0 m (9.0 m ²)	1.650 m	1.526 m	1.410 m
M	4.0 x 4.0 m (16.0 m ²)	1.650 m	2.064 m	1.910 m
L	4.0 x 6.0 m (24.0 m ²)	1.650 m	2.064 m	1.910 m

The model generates the node set of each size (four printed variants per size: foot, eave, ridge end and ridge middle, none handed) and exports each as STL, so R3 is met at model level. M and L share every tube length.

■ 4. Tube sections

Table 3. Section properties.

Tube	Area	Second moment	Section modulus	Mass
3/4 in EMT	86.7 mm ²	5,348 mm ⁴	457 mm ³	0.681 kg/m
1 in EMT	127.8 mm ²	12,639 mm ⁴	856 mm ³	1.003 kg/m

■ 5. Wind

At 20 m/s the dynamic pressure is $q = \frac{1}{2} \times 1.225 \times 20^2 = \mathbf{245 \text{ Pa}}$, giving 171.5 Pa of roof suction and 196.0 Pa on the windward wall.

A roof panel is 2.00 m (between frames) by 2.154 m (eave to ridge). With 45° tributary lines, each rafter edge takes a trapezoid of 1.154 m² (26.8 % of the panel) and the ridge and eave edges each take a triangle of 1.000 m². A wall panel is 2.00 m by 1.740 m; each post edge takes a triangle of 0.757 m² and the eave and ground edges trapezoids of 0.983 m². The TRL 2 estimate spread half of each panel uniformly onto the frames; the tributary rule gives a similar total but peaks at midspan, which raises the midspan moment by about 40 %.

Table 4. Member check at 20 m/s, middle frame unless stated.

Member	Tube	Moment	Stress	Factor on yield	Against 1.5
Rafter	1 in	141.8 N·m	166 MPa	1.66	Meets
Ridge tube	1 in	106.2 N·m	124 MPa	2.22	Meets (1.18 with 3/4 in in v0.1)
Post (windward)	3/4 in	86.0 N·m	188 MPa	1.46	Below
Eave tube, windward side	3/4 in	68.0 N·m	149 MPa	1.85	Meets
Eave tube, leeward side	3/4 in	80.9 N·m	177 MPa	1.55	Meets
End rafter, wind on the rear gable	1 in	134.3 N·m	157 MPa	1.75	Meets

Stress scales with the square of wind speed, so the rating is the speed at which the weakest member reaches a factor of 1.5.

Table 5. Wind rating and sensitivity.

Case	Rating at factor 1.5	Governing member
As decided: 1 in rafters and ridge tubes, 3/4 in posts and eave tubes (SNF-DDR-002 D8)	19.7 m/s (71 km/h, 44 mph)	Post, 1.46 at 20 m/s
TRL 2 concept, 3/4 in throughout	15.4 m/s	Rafter, 0.89 at 20 m/s
SNF-DDR-001 only: 1 in rafters, 3/4 in elsewhere (v0.1 baseline)	17.8 m/s	Ridge tube, 1.18 at 20 m/s
1 in rafters, ridge tubes and posts	20.3 m/s	Eave tube (leeward), 1.55 at 20 m/s
As decided, open gable facing the wind (internal pressure +0.55)	14.7 m/s (13.3 m/s in v0.1)	Eave tube (leeward), 0.81 at 20 m/s

For comparison, the TRL 2 method (half of each panel, uniform) gives 99 N·m and 218 MPa in a 3/4 in rafter, which matches SNF-PRC-001 v0.2; that method understated the midspan moment. The open gable case matters: with the door end facing the wind, internal pressure adds to the roof suction and the wall suction, and the rating falls to about 14.7 m/s. Whether the posts also move to 1 in waits for a frame analysis (SNF-DDR-002 D8); on this member-by-member basis 1 in posts would give 20.3 m/s. Closing the front gable (R7, still open) or dropping the skin in strong wind removes this case.

Snow is out of scope (SNF-DDR-001 D7). As a check for the safety note, 0.5 kPa of snow on plan puts 356 N·m and 416 MPa into a middle 1 in rafter, well past yield.

■ 6. Bracing, anchors and joints

Wind across the span puts 2,217 N on the two side walls; about 1,108 N reaches eave level. With the rear gable as the only braced frame across the span, its active X cable carries **1,209 N** and pulls up on its foot with 482 N. The roof acts as a cantilever from the rear gable; the resulting couple of 2,217 N·m puts 554 N into each side wall, 735 N into a side wall cable and another 482 N of uplift at its foot. Wind along the ridge on the closed rear gable (1,678 N) puts 556 N into a side wall cable. The highest cable tension is 1.2 kN, a factor of 6.6 on the assumed 8.0 kN breaking load of 4 mm wire rope; the hand tensioner rating is unknown.

Roof suction lifts the middle frame with 1,372 N against 116 N of its own weight, so each middle foot anchor sees **0.63 kN** and each end foot 0.31 kN. At a rear corner foot, the gable cable and the side wall cable both pull up as well: $0.31 + 0.48 + 0.48 = \mathbf{1.28\text{ kN}}$ before cable pretension, above the 1.0 kN target in R8.

At that foot the hitch pin bears on the polymer socket at 16.0 MPa, or **32.0 MPa** with the factor of 2 in R9, and on the EMT wall at 64 MPa. With the pin hole now 15 mm from the tube end (SNF-DDR-003 C1), the EMT shear-out capacity behind it is about 9.0 kN (37.8 kN at 50 mm in v0.3), seven times the load, so the steel is still not the limit; the polymer socket is. A rafter end transfers 198 N of shear into its socket, a nominal bearing pressure of 0.10 MPa over 65 mm of engagement.

The cables now clip to an M10 cable bolt through each eave and ridge node (SNF-DDR-003 C3). The worst node is a rear corner eave node, which takes the gable cable and a roof cable; taking both at full value together (1,765 N, an upper bound) with the ring 9.5 mm from the spot face gives 16.8 N·m and **171 MPa** in the shank, a factor of **2.6** on A4-70 stainless, and about 5.6 MPa of bearing on the polymer (11.2 MPa with the factor of 2 in R9). The printed tab it replaces would have carried about 46 MPa in bending at the gable cable load alone.

■ 7. Nodes, mass, packages and cost

Table 6. Printed nodes, size M, from the model volumes.

Node	Count	Solid volume	Printed mass	Cost each
Foot	6	697.3 cm ³	410 g	\$10.03
Eave (all six; corners with one blank socket)	6	434.3 cm ³	256 g	\$6.62
Ridge, end	2	419.3 cm ³	247 g	\$6.43
Ridge, middle	1	461.3 cm ³	271 g	\$6.97

The 15 nodes weigh 4.76 kg and cost about \$120 (v0.3: 4.97 kg, \$124; the printed tabs are gone and the finger recesses, bolt holes and second anchor slot remove a little more). With one eave node variant (SNF-DDR-002 D9), a size M kit prints four node variants in three families, none of them handed; the ridge nodes grow slightly because their ridge sockets now take 1 in tube.

Table 7. Mass and packages, size M.

Item	Mass
Tubes (33.74 m)	28.20 kg
Nodes	4.76 kg (v0.3: 4.97 kg)
Brace cables (29.5 m eye to eye), snap hooks and tensioners	3.62 kg (v0.3: 3.22 kg)
Cable bolt sets (9)	1.17 kg (new)
Screw anchors	3.60 kg
Guy lines	0.28 kg
Buttons and hitch pins	0.72 kg
Straps and bag	1.00 kg
Frame kit	43.3 kg (96 lb) (v0.3: 42.0 kg)
Tarpaulins, agency stock	9.1 kg
With tarpaulins	52.5 kg (v0.3: 51.1 kg)

The tube bundle weighs **28.6 kg** and takes about 0.030 m³; the bag weighs 14.7 kg (23.9 kg with the tarpaulins) and holds about 0.057 m³ of parts before the tarpaulins. Splitting the tubes into two bundles (rafters and ridge tubes 16.5 kg; posts and eave tubes 12.1 kg) would keep every package under 25 kg but makes three packages. The complete kit with tarpaulins is now 2.5 kg over the 50 kg carry limit of R11 (1.1 kg in v0.3).

Table 8. Cost, size M.

Item	Cost
Tubes: 10 sticks of 3/4 in, 8 of 1 in	\$202.00
Nodes (15)	\$119.74 (v0.3: \$124.30)
Brace cables (10)	\$42.00 (v0.3: \$35.00)
Cable bolt sets (9)	\$23.40 (new)
Screw anchors (8)	\$32.00
Guy lines (2)	\$6.00
Buttons and hitch pins	\$24.00
Straps and bag	\$20.00
Frame kit	\$469.14 (v0.3: \$443.30)
Value-engineering target	\$445 (a control target, not a limit): \$24.14 over
Tarpaulins, agency stock	\$50.00
With tarpaulins	\$519.14

The kit buys 54.9 m of tube and uses 33.7 m, a 39 % offcut.

■ 8. Skin and erection time

Full enclosure needs about 51.4 m² (roof with 150 mm overhang 19.4 m², side walls 14.4 m², each gable 8.8 m²); roof, side walls and one gable need 42.6 m². Two 4 x 6 m tarpaulins give 48.0 m², so the front gable stays open.

Erection time is estimated at about **46 min** for two adults: 63 person-minutes of one-person tasks (36 snap joints at 20 s, 8 anchors at 2 min, 10 cables at 1.5 min, skin 20 min) shared by two, plus 14 min of two-person tasks (layout and raising three frames). This is an estimate only; R4 needs a timed trial. The erection order of SNF-DDR-003 C7 (frames slid onto the tube ends, anchors last) uses the same tasks, so the estimate is unchanged.

■ 9. Results against requirements

Table 9. Requirement status, size M. Not met items first.

ID	Requirement	Value	Target	Status
R5	Ship flat	Longest member 2.064 m; tube bundle 28.6 kg, 0.030 m ³ ; bag 23.9 kg with tarpaulins	2.1 m or less; two packages, each 25 kg or less and 0.10 m ³ or less	Not met (tube bundle 3.6 kg over)
R6	Resist wind	Rating 19.7 m/s; post factor 1.46 at 20 m/s	Factor 1.5 at 20 m/s	Not met (posts; frame analysis pending)
R7	Fit standard tarpaulins	48.0 m ² closes roof, side walls and one gable (42.6 m ²); full enclosure 51.4 m ²	Both gables closed	Not met (front gable open; awaiting Amish)
R11	Carried by two people	Frame kit 43.3 kg; 52.5 kg with tarpaulins	50 kg or less	Not met (2.5 kg over with tarpaulins)
R13	Limit fire spread	Standard polyethylene tarpaulins	Flame spread test or fire-retardant option	Not met
R8	Anchor without a hammer	Demand up to 1.28 kN at a rear corner foot, before pretension	Each anchor holds 1.0 kN	At risk (demand above target; capacity not verifiable at TRL 3)
R9	Nodes strong in sun and cold	Pin bearing 32.0 MPa with the factor of 2	Factor 2 after two years, -10 to 70 °C	At risk (polymer not chosen; close to typical printed ASA strength at room temperature, lower at 70 °C)
R4	Go up without tools	About 46 min estimated	60 min or less	Not verifiable at TRL 3
R1	Living space	16.0 m ² (4.6 people at 3.5 m ²)	16 m ² or more	Met
R2	Headroom	73 % of floor at 2.0 m or more	60 % or more	Met
R3	Several sizes	S, M and L node sets generated and exported	Three sizes from one model	Met
R10	Value-engineering target	Frame kit \$469	\$445 value-engineering target, frame kit without tarpaulins	Over the value-engineering target by \$24.14 (reported against the target, not as met or not met)
R12	Repairable in the field	4 tube types; 3 node types in 4 printed variants, none handed	4 tube types and 3 node types	Met (at risk in v0.1)

Summary: 4 met, 5 not met, 2 at risk, 1 not verifiable at TRL 3, and R10 reported against the value-engineering target, \$24.14 over (v0.3: 5 met with R10 within the \$445 budget; v0.2: 4 met, 6 not met, 2 at risk, 1 not verifiable; v0.1: 4 met, 5 not met, 3 at risk, 1 not verifiable).

■ 10. Limitations

- Pressure coefficients are first-order, not taken from a code for this building shape and exposure. Gust factor, terrain and shelter from neighboring structures are not modeled.
- The frame is checked member by member. No frame analysis with pinned nodes and tension-only cables was run, so buckling of posts under combined axial load and bending, cable slack and second-order sway are not covered. That analysis is the next TRL 3 step under SNF-DDR-002 D8, before deciding on 1 in posts.
- Tarpaulin membrane forces that pull the edge members inward are ignored.
- EMT yield, anchor holding capacity, tensioner, ring and snap hook ratings and printed polymer strength at 70 °C are assumed or unknown.
- Costs are indicative and exclude labor, shipping and tooling.